

MATH 127 Calculus for the Sciences

Lecture 21



Today's lecture

Last time

IVT, discontinuities.

This time

Course note coverage Section 3.1.5 and 3.2.1

Limit at infinity

Indeterminant form

Limit at infinity

We have seen taking a limit of a function as x goes to a for some number a :

$$\lim_{x \rightarrow a} f(x).$$

This gives you the behaviour of $f(x)$ near $x = a$.

You can also take limit to infinity, which tells the behaviour at infinity. You might know this as horizontal asymptotes:

$$\lim_{x \rightarrow \infty} f(x),$$

$$\lim_{x \rightarrow -\infty} f(x).$$

Remark What are $f(\infty)$ and $f(-\infty)$?

Limit at infinity

Example What is

$$\lim_{x \rightarrow \infty} \frac{1}{x}$$

Limit at infinity

Example What is

$$\lim_{x \rightarrow -\infty} x^3$$

Limit at infinity

Example What is

$$\lim_{x \rightarrow \infty} e^{1/x}/x$$

Limit at infinity

Example What is

$$\lim_{x \rightarrow \infty} \frac{2x^3 + x}{x^3 + 1}$$

Limit at infinity

Example What is

$$\lim_{x \rightarrow \infty} \sin(x)$$

Determinate form

Recall we could have a determinate form

$$\lim_{x \rightarrow 0^+} \frac{1}{x} \rightsquigarrow \frac{1}{0^+}$$

This is not saying the right hand side is $\frac{1}{0}$, because that doesn't make sense. It is a notation to denote that we are dividing a constant number by a super small **positive** number, so the result *must be* ∞ .

This is called a **determinate form**. Another determinate form is $\frac{0}{\infty}$ which yields 0 because you divide a small number by a big positive number.

Question Can you come up with a determinate form that yields $-\infty$?

Indeterminate form

We could also have

$$\lim_{x \rightarrow -\infty} \frac{x+1}{x} \rightsquigarrow \frac{-\infty}{-\infty}$$

Again, this is not saying the right hand side is $\frac{-\infty}{-\infty}$, because ∞ is not a number and doesn't make sense to divide them. It says we are dividing a big **negative** number by a big **negative** number. **But** we do not know which one is bigger:

- If the numerator is 2 times bigger than the denominator, then the result is 2.
- If the numerator is half the denominator, then the result is $\frac{1}{2}$.

This is called a **indeterminate form** because we can not conclude what the answer must be. Another indeterminate form is $\frac{0^+}{0^+}$.

Question Can you come up with another indeterminate form?

Indeterminate form

Example Which one of the following is determinate, and which one is indeterminate?

- $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$
- $\lim_{x \rightarrow 0} \frac{\ln x}{x}$
- $\lim_{x \rightarrow \infty} \frac{e^x}{x^2 + 1}$
- $\lim_{x \rightarrow 0} \frac{\sin(x)}{2x}$
- $\lim_{x \rightarrow 1^-} \frac{e^x}{x - 1}$